Tripartite Entanglement in the HaPPY Code is not holographic

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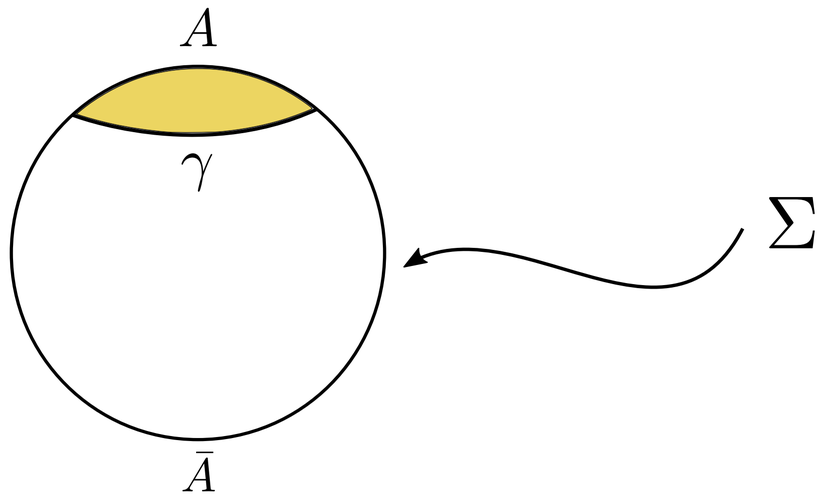
Question

Does a state $|\psi\rangle \in \mathscr{H}_{CFT}$ admit a smooth holographic dual?

- If yes, then it satisfies the Ryu-Takayanagi formula:

(S. Ryu, T. Takayanagi, *PRL* **96** 181602, arXiv: hep-th/0603001)

$$S(A) = \min_{\gamma} \frac{\mathscr{A}(\gamma)}{4G_N} .$$



Upshot: Holographic entropic inequalities.

Examples

Strong Subadditivity

(M. Headrick, T. Takayanagi, *PRD* 76 106013, arXiv: 0704.3719)

$$S(AB) + S(BC) \geq S(ABC) + S(B)$$

Negative Tripartite Mutual Information

(P. Hayden et al., *PRD* **87** 046003, arXiv: 1107.2940)

$$S(A) + S(B) + S(C) - S(AB) - S(BC) - S(AC) + S(ABC) \leq 0$$

The holographic entropy cone systematically generates many such inequalities.

(N. Bao et al., *JHEP* 09 130, arXiv:1505.07839)

Drawback

- Probes only bipartite features of holographic entanglement.
- For example, cannot distinguish

$$|W\rangle = \frac{1}{\sqrt{3}}|001\rangle + \frac{1}{\sqrt{3}}|010\rangle + \frac{1}{\sqrt{3}}|100\rangle$$

$$|aGHZ\rangle = \sqrt{\frac{2}{3}}|000\rangle + \frac{1}{\sqrt{3}}|111\rangle$$

(M. A. Nielsen, J. Kempe, *PRL* **86** 5184, arXiv: quant-ph/0011117)

Can we do better with multipartite entanglement measures?

Main Result

- Holographic states satisfy the GHZ-forbidding inequality:

(V. Balasubramanian et al., arXiv: 2509.03621.)

$$\frac{1}{2}R^{(3)}(A:B) \geq GM^{(3)}(A:B:C)\,.$$

Tripartite measures

- In stark contrast, stabilizer states satisfy

(S. Akella, arXiv: 2510.08520)

$$\frac{1}{2}R^{(3)}(A:B) \leq GM^{(3)}(A:B:C)\,.$$

Consequence (F. Pastawski et al., *JHEP 06, 149*, arXiv: 1503.06237)

Tripartite entanglement in the HaPPY code is not holographic.

Cannot come to this conclusion with entanglement entropies alone!

GHZ-Forbidding Inequality

- The LHS of the inequality is the residual information or the Markov gap.

(V. Balasubramanian et al., *JHEP* **06** 068, arXiv: 2411.03422) (P. Hayden et al., *JHEP* **10** 047 arXiv: 2107.00009)

$$R^{(3)}(A:B) = S^{(R)}(A:B) - I(A:B)$$

Reflected entropy

(S. Dutta and T. Faulkner, *JHEP* **03** 178, arXiv: 1905.00577)

- The RHS is the tripartite genuine multi-entropy.

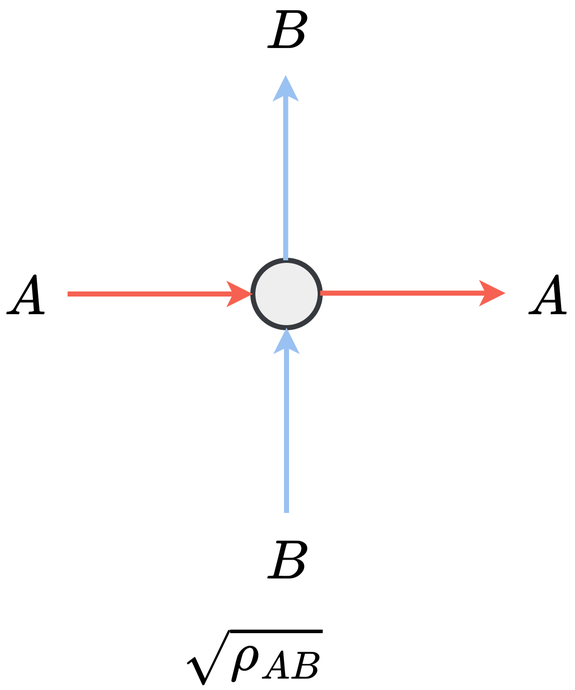
(N. Iizuka and M. Nishida, *PRD* **112** 026011, arXiv: 2502.07995)

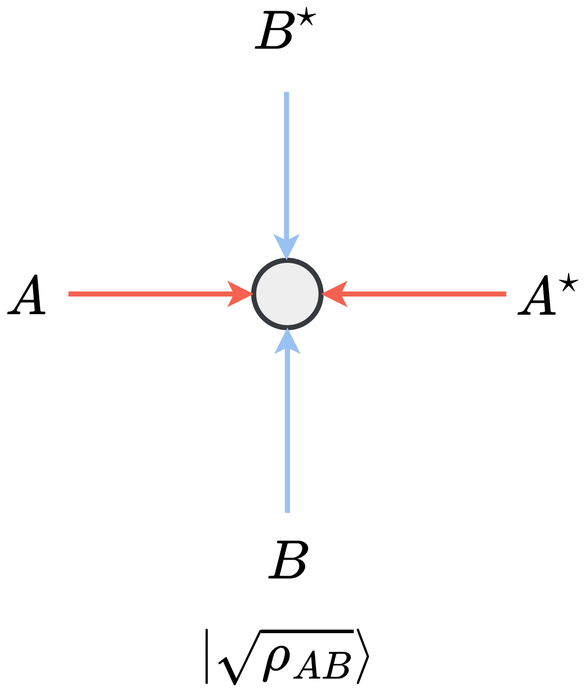
$$GM^{(3)}(A:B:C) = S^{(3)}(A:B:C) - \frac{1}{2}(S(A)+S(B)+S(C))\,.$$

Tripartite multi-entropy

(A. Gadde et al., *PRD* **106** 126001, arXiv: 2206.09723)

Reflected Entropy





$$\sqrt{\rho_{AB}} : \mathcal{H}_A \otimes \mathcal{H}_B \to \mathcal{H}_A \otimes \mathcal{H}_B$$

Canonical Purification

$$|\sqrt{\rho_{AB}}\rangle : \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_A^\star \otimes \mathcal{H}_B^\star \to \mathbb{C}\,.$$

Examples of Canonical Purification

Thermofield Double State.

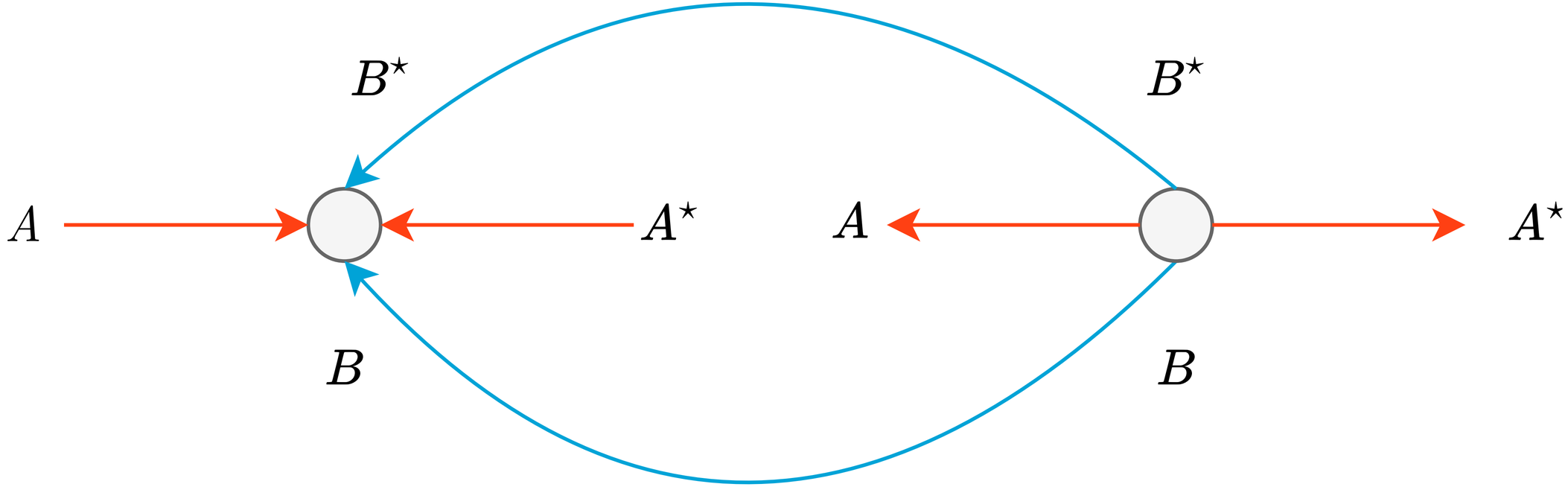
(J. Maldacena, *JHEP* **04** 021 arXiv: hep-th/0106112)

$$\rho = \frac{1}{Z(\beta)} \sum_E e^{-\beta E} |E\rangle\langle E| \longrightarrow |TFD\rangle = \frac{1}{\sqrt{Z(\beta)}} \sum_E e^{-\frac{\beta E}{2}} |E\rangle \otimes |E\rangle .$$

Mixed two-qubit state

$$\sqrt{\rho_{AB}} = \frac{1}{\sqrt{2}} |00\rangle\langle 00| + \frac{1}{\sqrt{2}} |11\rangle\langle 11| \longrightarrow |\sqrt{\rho_{AB}}\rangle = \frac{1}{\sqrt{2}} |0000\rangle + \frac{1}{\sqrt{2}} |1111\rangle$$

Reverse the bras into kets.



$$\rho_{AA^\star} = \mathrm{Tr}_{BB^\star} |\sqrt{\rho_{AB}}\rangle\langle\sqrt{\rho_{AB}}|$$

Mixed state $\mathscr{H}_A \otimes \mathscr{H}_{A^\star} \to \mathscr{H}_A \otimes \mathscr{H}_{A^\star}$

Reflected entropy: $S^{(R)}(A:B) = -\,\mathrm{Tr}_{AA^\star}\left(\rho_{AA^\star} \log \rho_{AA^\star}\right).$

Properties

- Bounded above and below by

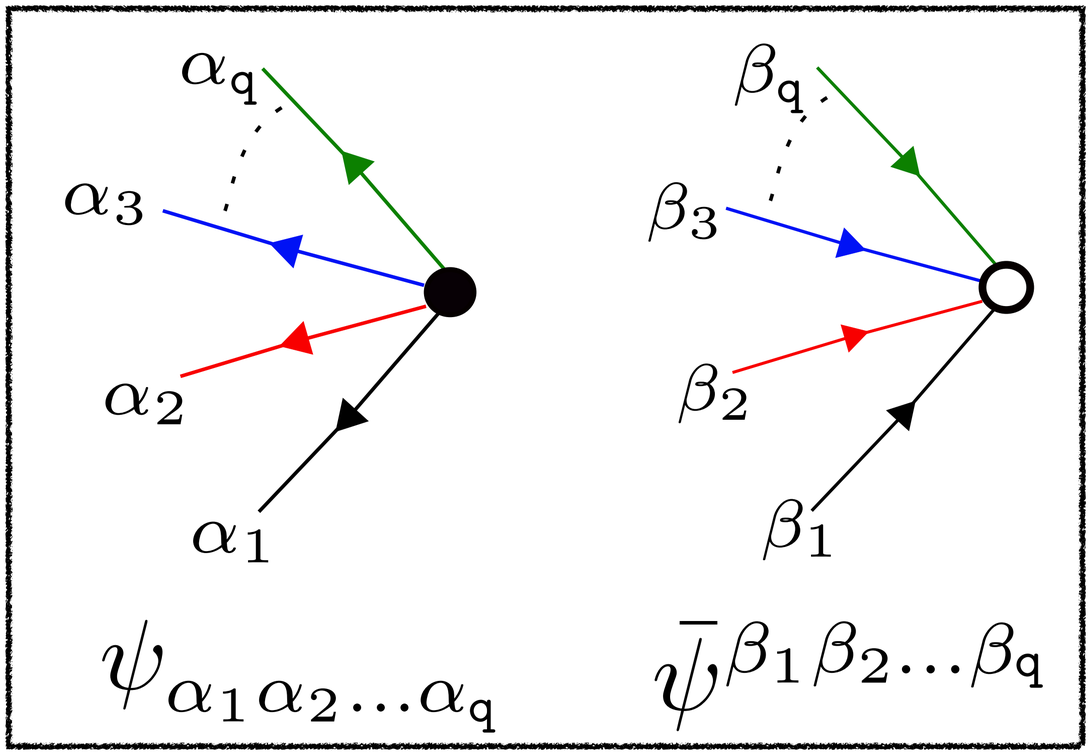
$$I(A:B) \leq S^{(R)}(A:B) \leq 2\min\{S(A), S(B)\}\,.$$

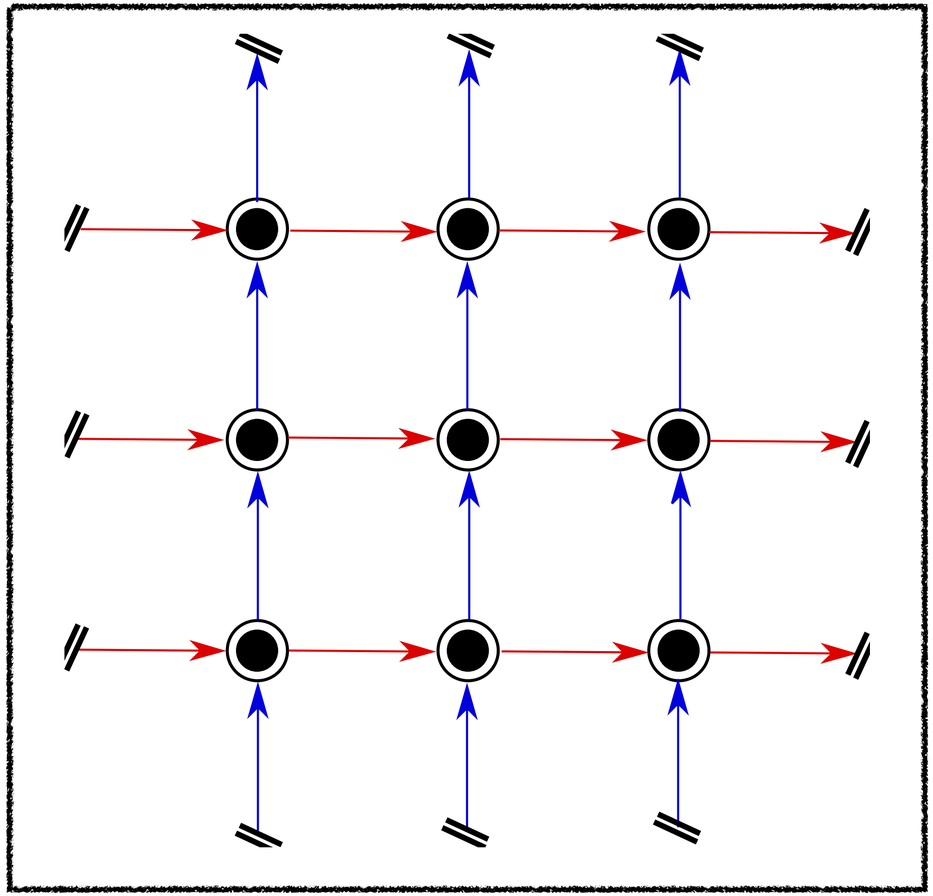
- The residual information (Markov gap) is the difference:

$$R^{(3)}(A:B) = S^{(R)}(A:B) - I(A:B)$$

- Zero for purely bipartite states and GHZ. In fact, zero for stabilizer states.

Genuine multi-entropy





$$|\psi\rangle = \sum_{\alpha_1=1}^{d_1} \ldots \sum_{\alpha_q=1}^{d_q} \psi_{\alpha_1\ldots\alpha_q} |\alpha_1\rangle \otimes \ldots \otimes |\alpha_q\rangle .$$

$$\sigma_a(\vec{r}) = (r_1, \ldots, r_a + 1, \ldots, r_q)$$

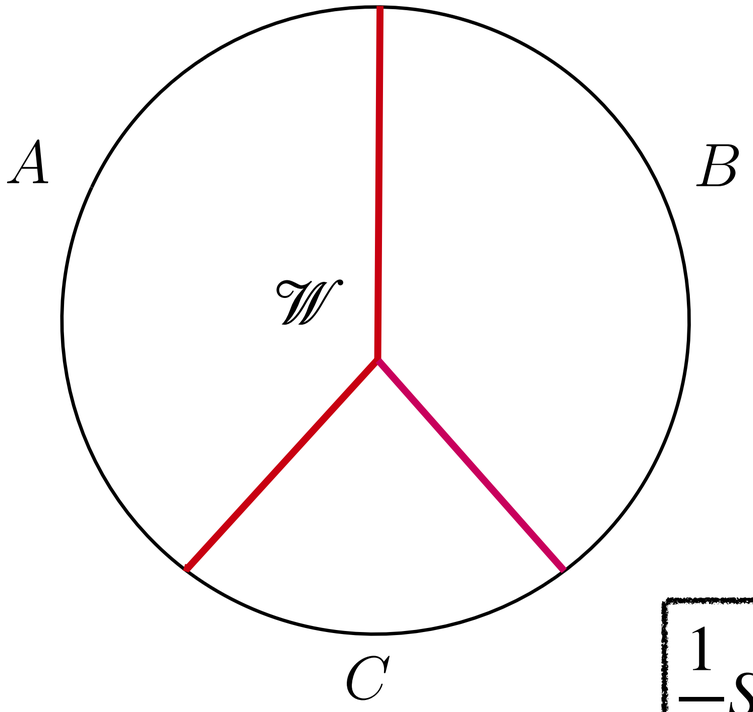
Multi-entropy

$$S^{(q)} = \lim_{n \to 1} \frac{1}{n(1-n)} \log \mathcal{Z}_n^{(q)}$$

Properties

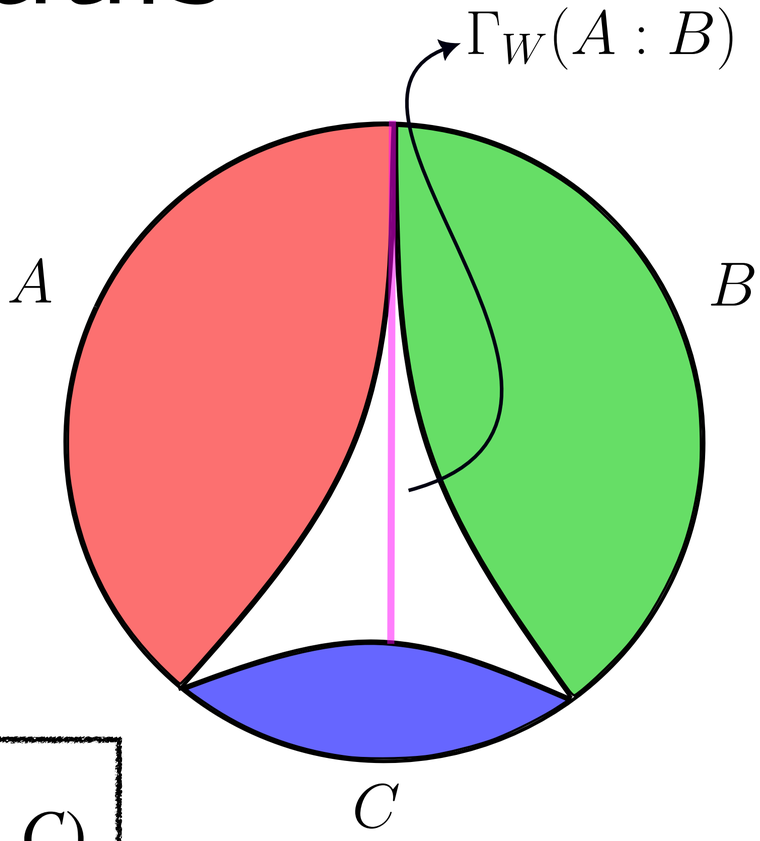
- Local unitary invariance.
- Symmetric in all parties.
- Generalization of entanglement entropy.

Holographic Duals



GHZ-forbidding inequality is straightforward.

$$\frac{1}{2}S^{(R)}(A:B) + S(C) \geq S^{(3)}(A:B:C)\,.$$



$$S^{(3)} = \frac{1}{4G_N}\mathscr{A}(\mathscr{W})$$

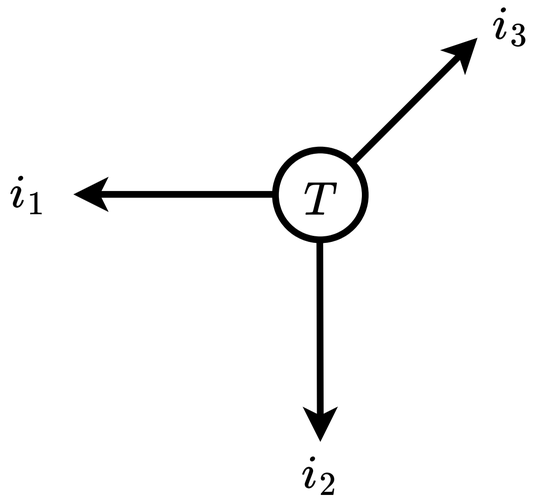
$$\frac{1}{2}S^{(R)}(A:B) = \frac{1}{4G_N}\mathscr{A}(\Gamma_W(A:B))\,.$$

HaPPY Code

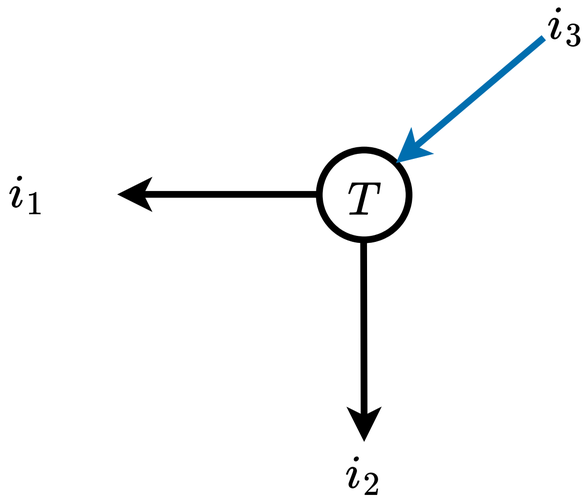
Perfect Tensors

Tensor with k legs: $T_{i_1\ldots,i_k}$.

$$|T\rangle = \sum_{i_1,\ldots,i_k} T_{i_1\ldots i_k} |i_1\ldots i_k\rangle \quad \Rightarrow \quad \sum_{i_1,\ldots,i_k} T_{i_1\ldots i_k} |i_1\ldots i_{k-1}\rangle\langle i_k|$$



$T_{i_1 i_2 i_3} |i_1 i_2 i_3\rangle$



$T_{i_1 i_2 i_3} |i_1 i_2\rangle\langle i_3|$

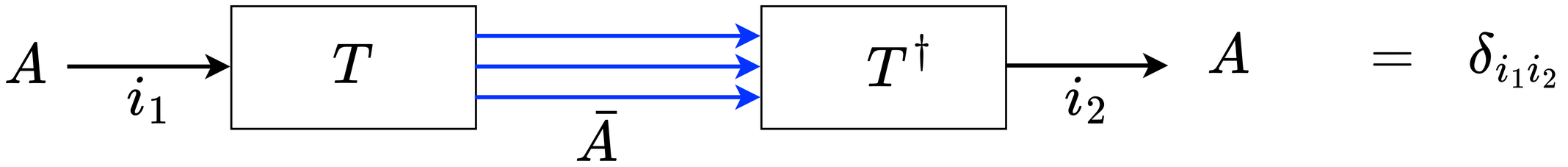
Every tensor is a map from a subset of legs to its complement.

Even-legged tensor. Partition its legs into A and $\bar{A}$, with $|A| \leq |\bar{A}|$.

$$T : \mathscr{H}_A \to \mathscr{H}_{\bar{A}}$$

Perfect tensor:

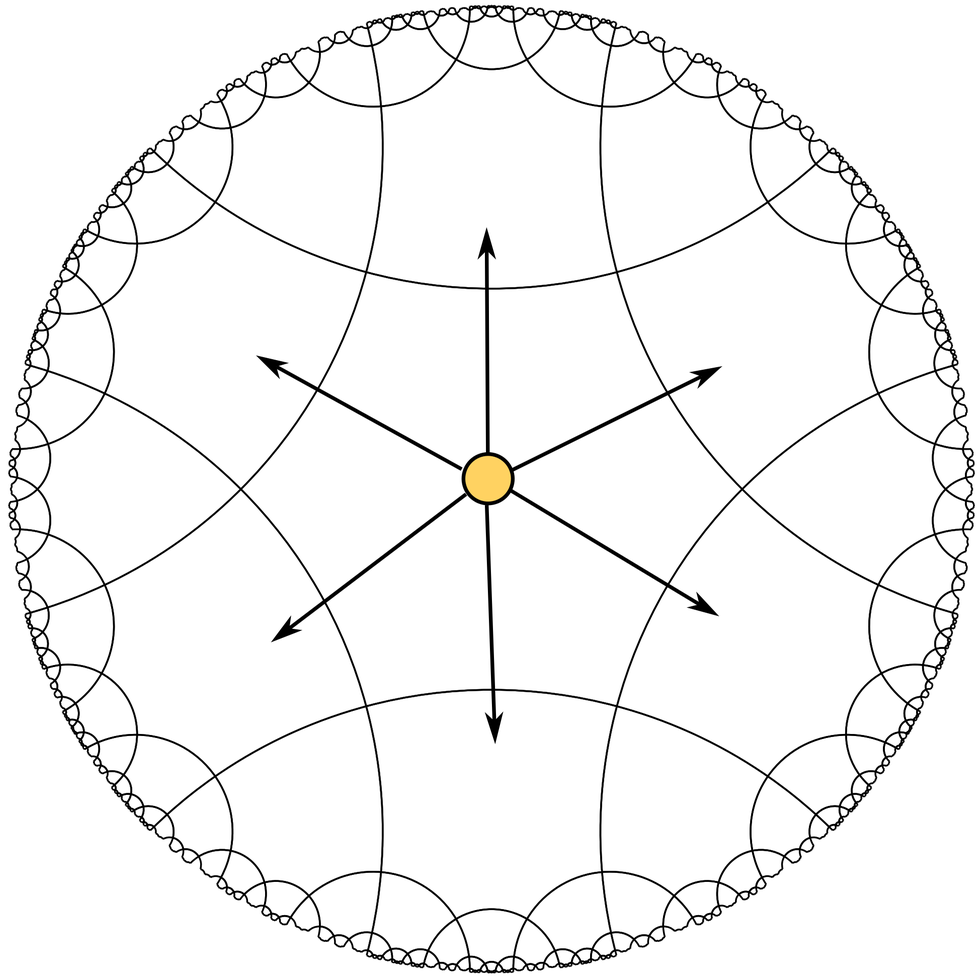
$$T^\dagger T = \mathbf{1}_A \qquad \text{(Isometry for all partitions)}$$



Every perfect tensor is an *absolutely maximally entangled (*AME) state.

(W. Helwig and W. Cui, arXiv: 1306.2536)

Building the Network



1. Place a 6-legged perfect tensor, e.g. AME(6, 2), at each hexagon.

2. Contract all bulk legs to prepare a boundary state.

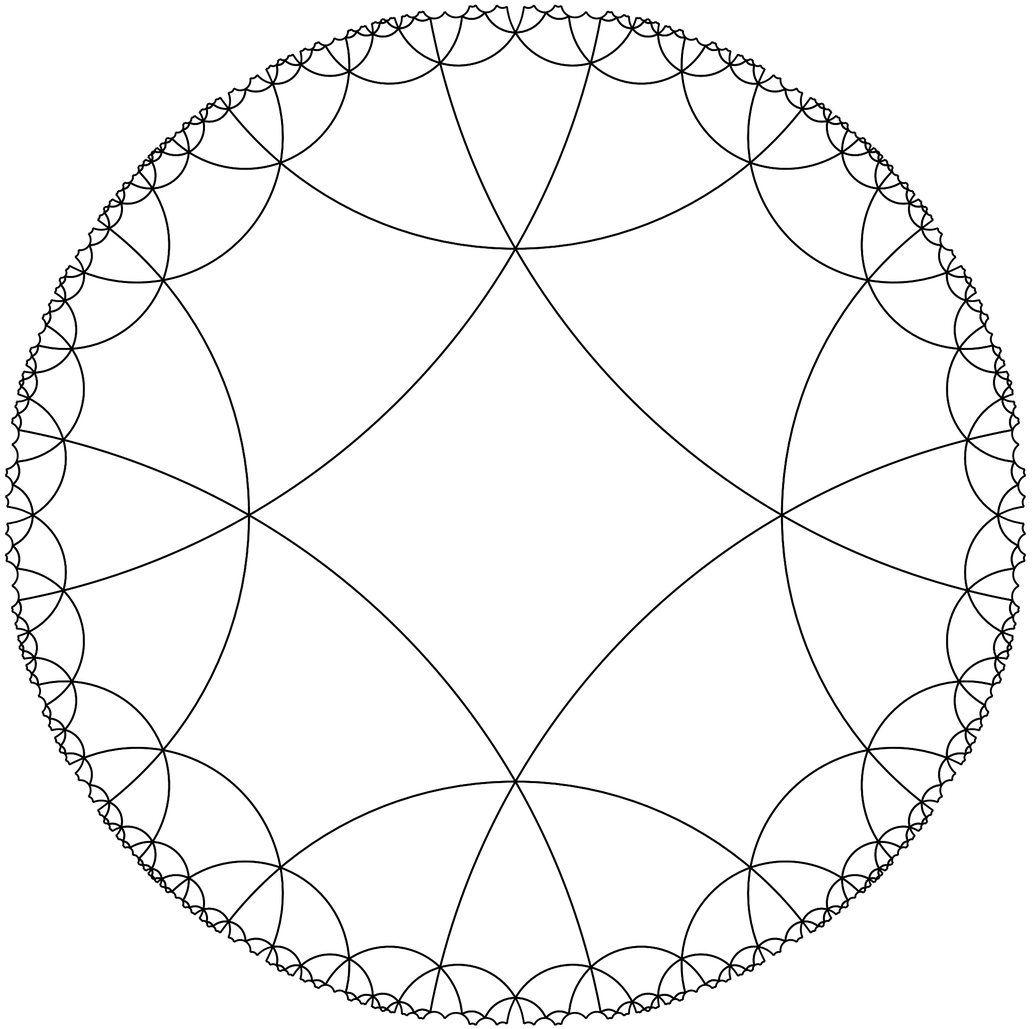
3. Satisfies all inequalities coming from the entropy cone.

4. Violates GHZ-forbidding inequality.

Non-Stabilizer AME States

- Most AME states are stabilizer states.
- There are non-stabilizer AME states, e.g. AME(4, 6).

  (S. A. Rather et al., *PRL* **128** 080507, arXiv: 2104.05122)
- May satisfy GHZ-forbidding inequality.



Stabilizer Formalism

Stabilizer States

Definition: An n-qubit state $|\psi\rangle$ stabilized by n independent commuting elements $\{g_1, \ldots, g_n\}$ of the n-qubit Pauli group $\mathscr{P}_n$.

Generators

Element of $\mathscr{P}_n$ $\longrightarrow$ $\mathbf{P} = i^k P_1 \otimes \ldots \otimes P_n$ $\qquad k = 0,1,2,3$ $\qquad P_i = I, X, Y, Z$

$$g_i |\psi\rangle = |\psi\rangle$$

$$[g_i, g_j] = 0$$

Stabilizer group: $G = \langle g_1, \ldots, g_n \rangle \cong \mathbb{Z}_2^n$

Examples

Bell state

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Stabilized by $g_1 = XX$ and $g_2 = ZZ$.

GHZ State

$$|GHZ\rangle = \frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle\,.$$

Stabilized by $g_1 = XXX$, $g_2 = ZIZ$, and $g_3 = IZZ$.

Non-Example

W-State

$$|W\rangle = \frac{1}{\sqrt{3}}|001\rangle + \frac{1}{\sqrt{3}}|010\rangle + \frac{1}{\sqrt{3}}|100\rangle$$

Not stabilized by any Pauli string except identity.

Tripartite entanglement in stabilizer states

Theorem (GHZ extraction). (S. Bravyi et al., *Jour. Math. Phys.* **47** (2006), arXiv: quant-ph/0504208)

Any tripartite stabilizer state is local-unitary equivalent to a collection of: (a) GHZ states, (b) Bell pairs, and (c) unentangled qubits.

Number of GHZs extracted

$$p = \log_2 \left(\frac{|G|}{|G_{AB} \cdot G_{BC} \cdot G_{AC}|} \right)$$

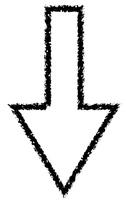
Number of Bell pairs extracted between A and B

$$m_{AB} = \frac{1}{2} \left(\log_2 \left(\frac{|G_{AB}|}{|G_A||G_B|} \right) - p \right)$$

Using this theorem, we find the following expressions for the reflected entropy and the tripartite multi-entropy:

$$S^{(R)}(A:B) = \log_2\left(\frac{|G_{AB}|}{|G_A||G_B|}\right)$$

$$S^{(3)}(A:B:C) = \frac{1}{2}\log_2\left(\frac{|G|}{|G_A||G_B||G_C|}\right) + \frac{1}{2}\log_2\left(\frac{|G|}{|G_{AB}\cdot G_{BC}\cdot G_{AC}|}\right).$$

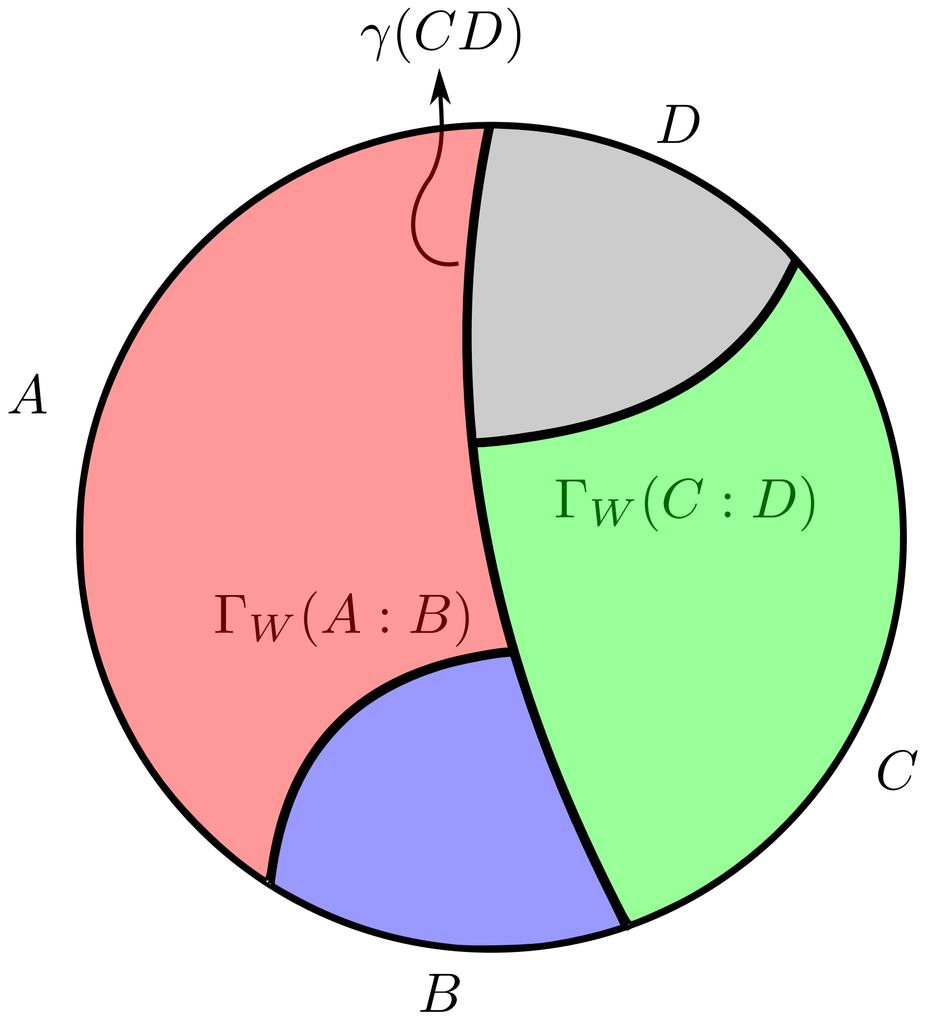


GHZ-forbidding inequality becomes

$$|G_{AB}\cdot G_{BC}\cdot G_{AC}| \geq |G| \quad (!?)$$

Thoughts for the future

Four (and more) parties



$$\frac{1}{2}S^{(R)}(A:B)+\frac{1}{2}S^{(R)}(C:D)+S(CD)\geq S^{(4)}(A:B:C:D)\,.$$

Is there a systematic way to generate multiparty holographic inequalities?

Multipartite version of the entropy cone program?

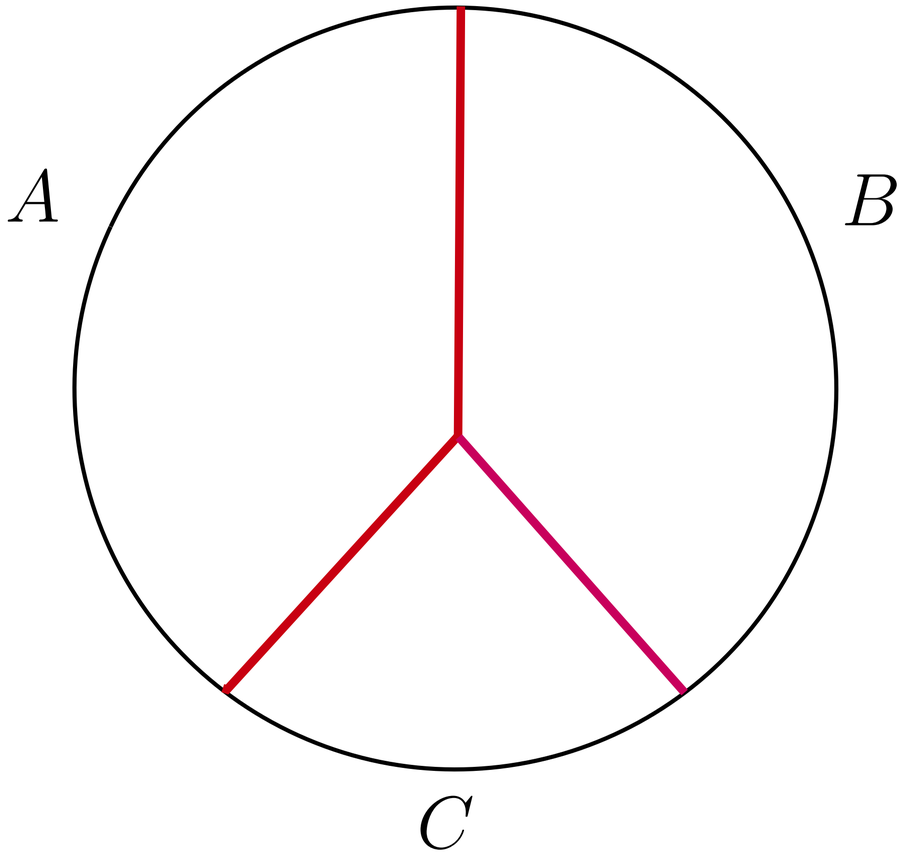
Random Tensor Networks

Do random tensor networks satisfy the GHZ-forbidding inequality?

- Reflected entropy in random tensor networks agrees with expectations from holography.

(C. Akers et al., *JHEP* **05** (2022) 162, arXiv:2112.09122)

Does multi-entropy in random tensor networks reproduce the Mercedes-Benz?



Take-Home Message

- The HaPPY code satisfies all inequalities coming from the entropy cone, but violates the GHZ-forbidding inequality.
- Powerful probe of multipartite holographic entanglement.
- Excludes stabilizer states as candidate holographic states.